Forward Substitution

$Lx = b$

↓

Lower triangular matrix.

$$x = -1$$
$$2x + y = 0$$
$$3x + 4y + 2 = 1$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \qquad b = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$Lx = b$

$x = -1$

$2x + y = 0$

$2(-1) + y = 0$

$y = 2$

$3x - 1 + 4 \times 2 + z = 1$

$-3 + 8 + z = 1$

$z = -4$

$$x = \begin{bmatrix} -1 \\ 2 \\ -4 \end{bmatrix}$$

Backward

Backward Substitution.

Use to solve the system

$Ux = b$

where U is an upper triangular matrix

$$U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

Ex :-

$$3x_1 + 2x_2 + x_3 = b$$
$$2x_2 + 3x_3 = 7$$
$$2x_3 = 4$$

$$U = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{pmatrix} \qquad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$b = \begin{pmatrix} b \\ 7 \\ 4 \end{pmatrix}$$

$$Ux = b$$

① $2x_3 = 4$

$x_3 = 2.$

② $2x_2 + 3x_3 = 7$

$2x_2 + 3x2 = 7$

$2x_2 = 7 - b$

$x_2 = 0.5$

③ $3x_1 + 2x_2 + x_3 = b$

$3x_1 + 2x0.5 + 2 = b$

$3x_1 = b - 3$

$x_1 = 1$

CamScanner

Gaussian Elimination Method.

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + 2x_2 + 4x_3 = -1$$

$$x_1 + 3x_2 + 9x_3 = 1$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

~~Change as~~

$$\begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \xrightarrow{R_2 - R_1 \to R_2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & -2 \\ 1 & 3 & 9 & 1 \end{bmatrix}$$

$$\downarrow \; R_3 - R_1 \to R_3$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & -2 \\ 0 & 2 & 8 & 0 \end{bmatrix} \xrightarrow{R_3 - 2R_2 \to R_3} \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$\downarrow$

This is upper triangular matrix.
So we can use Backward substitution.

CamScanner